I. PLC Data Type Reference (Future)

Appendix I of the EFC Gases & Advanced Materials Controls Standard, Revision I, October 14, 2026. Standalone copy; see the full document for the other chapters and appendices.

[Future: This appendix describes the proposed Process Control Library (PCLib v1.00), the next generation of EFC standard data types. It has not been adopted yet. Existing systems keep using the current types in Appendix H of the Controls Standard until a project is approved to move to PCLib. Member names, initial values, and alarm settings may still change during review and simulator testing.]

PCLib is a set of standard data types (Control Expert DDTs and matching Ignition UDTs) for process-control equipment. Compared with the current library it gives every piece of equipment a consistent structure: the same member prefixes, the same mode and alarm objects, the same status integers for HMI animation, and a documented link to the industry standards the types are aligned with (ISA-88, ISA-18.2, ISA-101, NAMUR NE107/NE43, and MTP). The library is kept in the EFC Repo under `PCLib/`. The type definitions are generated by `gen_pclib.py`, which writes both the Control Expert import file and the Ignition import file, so the PLC and SCADA definitions always match member for member. Change the generator and re-run it; never hand-edit the generated files.

The types are **data only**. Each type is paired with a DFB (or ST section) that holds its behaviour; most are laid out so the DFB can take the whole DDT as a single `IN_OUT` parameter. The first of these, `FB_AI`, is summarized in Section I.13.

I.1 Current to Future Mapping

The table shows which PCLib type replaces each current type. Members do not map one to one, so moving an existing project to PCLib means re-mapping its I/O, logic, and SCADA bindings, and is done as a planned change (Section 7.3 of the Controls Standard), not as a find-and-replace.

Table I.1: Current to PCLib Type Mapping

Current type	PCLib type	Notes
`DDT_Process_Value`, `AI_Sensors`	`T_AI`	Scaling, filtering, NE43 range check, quality, and HH/H/L/LL/ROC alarms in one type.
`DDT_AI_Alarm`	`T_AlmLim`	One object per limit, with deadband, delays, shelving, and the ISA-18.2 state.
`DDT_Discrete`	`T_DI`	Override becomes simulation (`Sim_`); optional state alarm.
`DDT_Valve`	`T_DValve`	Single or dual solenoid, limit switches, travel alarms, run counters.
`DDT_Motor_VFD`	`T_VFD` (or `T_Motor` for fixed speed)	Fixed-speed and variable-speed drives become separate types.

Table I.1 continued

Current type	PCLib type	Notes
DDT_Device	T_Mode + T_Alarm	Mode, ownership, and alarms become nested objects used by every device type.
DDT_Setpoint	Cfg_ / Prm_ members	Setpoints live inside the type that uses them (for example T_PID, T_AlmLim). Hard-coded clamp limits (Section 10.3.2 of the Controls Standard) stay in the DFB.
DDT_Scale	T_AI (weight) + T_Totalizer	Network-connected scales still need an application-specific wrapper *TBD*.
DDT_Seq	T_Phase	ISA-88 Part 1 state model and commands replace the custom step/status words.
DDT_Station	T_Unit + T_Phase	Application-specific; a station demo type (DDT_Station_Demo) is in the EFC Repo for review.
PDC compressor package interface (project-specific)	T_DiaComp	Two-stage diaphragm compressor unit with leak detection, oil systems, purge, and first-out (Section 10.9.1 of the Controls Standard). The current project type will be mapped once it is reviewed *TBD*.
(no current type)	T_Intlk, T_ESD	Interlock and shutdown groups with first-out and timed bypass (BPCS layer only).
(no current type)	T_PID, T_CValve, rotating and process equipment types	New coverage; see the type index.

I.2 Naming Conventions

Every member name starts with a prefix that says what it is and who writes it. Identifiers follow IEC 61131-3 and stay within the 32-character Control Expert limit.

Table I.2: PCLib Member Prefixes

Prefix	Meaning	Written by
Inp_	Field / upstream input	I/O mapping or upstream logic
Cfg_	Configuration / tuning	Engineer (HMI with access control)
Cmd_	Operator command. Booleans are **one-shot**: the logic clears them after acting	HMI
Prog_	Command from the ISA-88 owner (phase, equipment module, batch server)	Owner logic
Prm_ / Rpt_	ISA-88 recipe parameters / report values	Batch server / phase logic
Out_	Output to field or downstream logic	Equipment logic
Sts_	Status	Equipment logic
Val_	Calculated value	Equipment logic
Alm_	Alarm (T_Alarm / T_AlmLim)	Equipment logic sets Inp_Cond
Sim_	Simulation	Engineer

- Booleans ending in OK are **TRUE = healthy**, so a broken wire or lost signal reads as unhealthy. Other Inp_ booleans are TRUE when the condition is present.
- Interlocks (Inp_IntlkOK) stop running equipment. Permissives (Inp_PermOK) only block a start.

- `Sts_State` / `Sts_Mode` / `Sts_NE107` integers are for HMI animation. The values are listed in each member comment.
- These are **data types only**. Pair each one with a DFB (or ST section) holding the behaviour. Most types are laid out so a DFB can take the whole DDT as one IN_OUT parameter.

[`T_ESD`, `T_Intlk`, and `T_GasDet` are for the basic process control layer and the HMI only. Safety instrumented functions with a SIL target must run in a certified safety controller (for example M580 Safety) and follow IEC 61511.]

I.3 Standards Alignment

The standards define models, behaviour, and terminology; apart from MTP they do not prescribe PLC data layouts. PCLib carries the data each model needs so the DFBs can implement the behaviour. This is alignment by design, not a third-party certification. Alarm priorities, limits, and equipment states still come from EFC's alarm philosophy and rationalisation (Section 5.7 of the Controls Standard).

Table I.3: PCLib Standards Alignment

Standard	Scope	How PCLib applies it
ISA-88.00.01 / IEC 61512-1	Batch control: physical and procedural models, modes, states	Every type has an ISA-88 role (control module, equipment module, unit, phase). `Prog_` members are commands from the owning phase or equipment module. `T_Mode` adds arbitration (acquire / release / owner ID). `T_Phase` uses the Part 1 example state model (12 states) and commands (Start, Stop, Hold, Restart, Abort, Reset, Pause, Resume), with Automatic / Semi-automatic / Manual modes, recipe parameters and reports. `T_Unit` holds allocation, batch identity and equipment state (clean, dirty, in use...).
ISA-18.2 / IEC 62682	Alarm management	`T_Alarm` uses the standard's state model (A Normal to H Out-of-service), including latching, time-limited shelving, suppressed-by-design and out-of-service. Chattering control (deadband, on/off delay), priority and class come from rationalisation. Events that need no operator action are status bits, not alarms (e.g. pump changeover).
ISA-101.01	HMI design	Modes, states and NE107 status are integers, so graphics can use consistent symbols without colour logic in the PLC. `Sts_CmdMask` lets the HMI grey out invalid phase commands. Tag and description are in the type for faceplates.
ISA-5.1	Instrument identification	`Cfg_Tag` holds the ISA-5.1 tag (e.g. FT-101). Limit alarms follow HH / H / L / LL naming.
NAMUR NE107	Field device diagnostics	`Sts_NE107` on every field-device control module (Good / F / C / S / M). `Inp_NE107` takes the status from HART or fieldbus devices.
NAMUR NE43	4-20 mA failure signal levels	`T_AI` bad-signal limits default to NE43 (below 3.6 mA, above 21 mA). `T_GasDet` fault and over-range thresholds are configurable.
VDI/VDE/ NAMUR 2658-4 (MTP)	Module Type Package data objects for modular plants	Each type lists its nearest MTP data assembly (AnaMon, BinMon, MonBinVlv, MonAnaVlv, MonBinDrv, MonAnaDrv, PIDCtrl, LockView16). Mode values map to MTP Offline / Operator / Automatic. Member names are not MTP attribute names, so an MTP export needs a mapping layer.
ISA-84 / IEC 61511	Safety instrumented systems	`T_Intlk` and `T_ESD` follow the bypass-management requirements (authorisation, time limit, indication, logging data), first-out, manual reset after trip and de-energise-to-trip. They are **basic process control (BPCS) functions only**, not SIFs.

Table I.3 continued

Standard	Scope	How PCLib applies it
IEC 61131-3	PLC languages and data types	Elementary IEC data types only. Identifiers are checked against the 32-character limit and IEC / Control Expert reserved words.
ISA-95 / IEC 62264	Enterprise-control integration	Level 1-2 data. `T_Unit` and `T_Phase` expose batch ID, recipe, product and reports for MES collection.

I.3.1 ISA-88 Roles

Table I.4: PCLib Types by ISA-88 Level

ISA-88 level	PCLib types
Support	`T_Alarm`, `T_AlmLim`, `T_Mode`, `T_RunTime`, `T_Intlk`, `T_ESD`
Control module	`T_AI`, `T_DI`, `T_DValve`, `T_CValve`, `T_Motor`, `T_VFD`, `T_PID`, `T_PumpProt`, `T_Totalizer`, `T_HeatEx`, `T_Analyzer`, `T_GasDet`
Equipment module	`T_PumpDS`, `T_Agitator`, `T_Compressor`, `T_Conveyor`, `T_DosePump`
Equipment module / Unit vessel	`T_Tank`
Equipment phase	`T_Phase`
Unit	`T_Unit`

Control modules take `Prog_` commands only from the owner that has acquired them through `Mode.Prog_Acquire` (ISA-88 arbitration). Out-of-service and Maintenance modes are site extensions; ISA-88 itself defines Automatic, Semi-automatic and Manual.

I.4 Status Values

The status integers below are the same in every type that uses them, so one set of SCADA symbols and colours works across the library (ISA-101).

Table I.5: PCLib Status Values

Object	Member	Values
Alarm (`T_Alarm`, `T_AlmLim`)	`Sts_State`	0=A Normal 1=B Unack 2=C Acked 3=D RTN-Unack 4=E Latched 5=F Shelved 6=G Suppressed-by-design 7=H Out-of-service
Phase (`T_Phase`)	`Sts_State`	0=Idle 1=Running 2=Complete 3=Pausing 4=Paused 5=Holding 6=Held 7=Restarting 8=Stopping 9=Stopped 10=Aborting 11=Aborted
Phase commands	`Prog_Cmd`, `Sts_CmdMask`	1=Start 2=Stop 3=Hold 4=Restart 5=Abort 6=Reset 7=Pause 8=Resume
Equipment (`T_Unit`, `T_Tank`)	`Sts_EqState`	on `T_Unit` and `T_Tank`: 0=Available-clean 1=In use 2=Dirty 3=Cleaning (CIP) 4=Sterilising (SIP) 5=Maintenance 6=Out of service
Mode (`T_Mode`)	`Sts_Mode`	0=Out of service (L) 1=Maintenance (MT) 2=Manual (M) 3=Automatic (A). Definitions and allowed changes: Section 10.3.1 of the Controls Standard.

Table I.5 continued

Object	Member	Values
Field device health	`Sts_NE107`	NAMUR NE107: Good, F (failure), C (function check), S (out of specification), M (maintenance required)

I.5 Importing into Control Expert and Ignition

Control Expert Classic V16.2.

1. In the Project Browser, right-click **Derived Data Types**, choose **Import**, and select `export/PCLib_CE_V16.2.xdd` (all types in dependency order). If a type already exists, choose whether to keep or replace it.
2. Run **Build > Analyze Project**.
3. Declare one unlocated instance per piece of equipment, for example `P101 : T_Motor;` or `FT101 : T_AI;`.
4. For Ignition (or any OPC UA client) to read instances by name, enable the data dictionary: **Tools > Project Settings > PLC embedded data > Data dictionary**, then rebuild and download.

Totals use `LREAL`; if the CPU does not support it, regenerate with `gen_pclib.py --no-lreal`.

Ignition 8.1 Perspective. `export/PCLib_Ignition81.json` holds the matching UDT definitions in a `PCLib` folder. Nested DDTs (`Mode`, `Alm_*`, `Run`, `Drv`) become nested UDT instances, and arrays become folders with one tag per element.

1. In the Designer Tag Browser, select **Data Types**, choose **Import Tags**, and select the JSON file.
2. Create the OPC UA connection to the PLC (M580 embedded server, BMENUA0100, or OPC UA Server Expert; see Section 8.2 of the Controls Standard).
3. Create a UDT instance under **Tags** (for example `P101` of type `PCLib/T_Motor`) and set its parameters as in Table I.6.
4. Bind Perspective faceplates to the instance. `Cfg_`, `Cmd_`, `Sim_`, and `Prm_` tags are writable; everything else is read-only, so the HMI never writes PLC status (ISA-101).

Table I.6: PCLib Ignition UDT Parameters

Parameter	Meaning	Example
`OPCServer`	Ignition OPC connection name	`M580_Line1`
`PathPrefix`	Node-ID prefix used by your OPC server	`ns=2;s=` (check with the OPC Quick Client)
`BaseTag`	PLC instance name	`P101`

Each tag's OPC path is {`PathPrefix`}{`BaseTag`}.<`Member`>. Browse one variable in the OPC Quick Client first and copy its prefix, because the namespace index and path syntax differ between Schneider OPC servers.

Table I.7: PCLib Data Type Mapping

Control Expert	Ignition	Note
BOOL	Boolean	
INT	Int2	
UINT, WORD, DINT	Int4	
UDINT	Int8	
REAL / LREAL	Float4 / Float8	
TIME	Int4	milliseconds (`engUnit` = ms)
DT	DateTime	if your OPC server delivers DT as a number, change these tags to Int8
STRING[n]	String	

Alarms: each `T_Alarm` / `T_AlmLim` UDT has an Ignition alarm on `Sts_Active`, so PLC alarms appear in the Perspective Alarm Status Table and alarm journal. The alarm is enabled by `Cfg_En`, and its priority comes from `Cfg_Priority`. EFC uses Ignition's priority numbering (0=Diagnostic 1=Low 2=Medium 3=High 4=Critical, Table 11.1 of the Controls Standard) **NEW**.

[The member tables below still show the v1.00 numbering (1=Emergency to 4=Low). They are regenerated from `gen_pclib.py`, which will be updated to Ignition numbering at the next PCLib revision.]

Acknowledging in Ignition runs a tag event script that writes `Cmd_Ack` to the PLC, so the PLC alarm state (ISA-18.2) stays authoritative. Shelving is done in the PLC through `Cmd_Shelve`; Ignition's own shelving only hides the alarm in Ignition.

I.6 Type Index

Table I.8: PCLib Type Index

Type	ISA-88 role	MTP equivalent	Mem.	Depends on
T_Alarm	Support (alarm object)	Alarm/Warning/ Tolerance limit attributes	26	-
T_AlmLim	Support (alarm object)	e.g. VAHLim / VAHEn / VAHAct	28	-
T_Mode	Support (mode + ownership)	OpMode (Offline/ Operator/Automatic) + SrcMode	20	-
T_RunTime	Support (asset data)	-	11	-
T_AI	Control module	AnaMon	35	T_Alarm, T_AlmLim
T_DI	Control module	BinMon	19	T_Alarm
T_DValve	Control module	MonBinVlv	41	T_Alarm, T_Mode, T_RunTime
T_CValve	Control module	MonAnaVlv	42	T_Alarm, T_Mode
T_Motor	Control module	MonBinDrv	52	T_Alarm, T_AlmLim, T_Mode, T_RunTime
T_VFD	Control module	MonAnaDrv	82	T_Alarm, T_Mode, T_RunTime
T_PID	Control module	PIDCtrl	51	T_Alarm, T_AlmLim

Table I.8 continued

Type	ISA-88 role	MTP equivalent	Mem.	Depends on
T_PumpProt	Control module	-	28	T_Alarm, T_AlmLim
T_PumpDS	Equipment module	-	26	T_Alarm, T_Mode
T_Agitator	Equipment module	MonAnaDrv (drive)	15	T_Alarm, T_AlmLim, T_Mode, T_RunTime, T_VFD
T_Compressor	Equipment module	MonAnaDrv (drive)	29	T_Alarm, T_AlmLim, T_Mode, T_RunTime, T_VFD
T_Conveyor	Equipment module	MonBinDrv (drive)	28	T_Alarm, T_AlmLim, T_Mode, T_Motor, T_RunTime
T_DiaComp	Equipment module	MonBinDrv (drive)	89	T_Alarm, T_AlmLim, T_Mode, T_Motor
T_Tank	Equipment module / Unit vessel	-	41	T_Alarm, T_AlmLim
T_Totalizer	Control module	-	41	T_Alarm, T_AlmLim
T_HeatEx	Control module (monitoring)	-	31	T_AlmLim
T_DosePump	Equipment module	MonAnaDrv (drive)	31	T_Alarm, T_AlmLim, T_Mode, T_Motor, T_RunTime
T_Analyzer	Control module	AnaMon	28	T_AI, T_Alarm, T_AlmLim
T_GasDet	Control module	AnaMon	39	T_Alarm, T_AlmLim
T_Intlk	Support (interlock group)	LockView16	26	T_Alarm
T_ESD	Support (shutdown group)	-	34	T_Alarm
T_Phase	Equipment phase (procedural)	Service (state machine)	42	T_Alarm
T_Unit	Unit (physical)	-	26	T_Alarm

I.7 Common Building Blocks

I.7.1 T_Alarm

Discrete alarm, ISA-18.2 / IEC 62682 state model, shelving, suppression, out-of-service.

ISA-88 role: Support (alarm object) **MTP:** Alarm/Warning/Tolerance limit attributes **Standards:** ISA-18.2 / IEC 62682, ISA-101.01

Table I.9: T_Alarm Members

Member	Type	Init	Description
Inp_Cond	BOOL		Raw alarm condition (written by owner logic)
Inp_Suppress	BOOL		Suppressed-by-design: state-based suppression by logic (ISA-18.2 state G)
Cfg_En	BOOL	TRUE	Alarm configured (FALSE = not an alarm, e.g. instrument not fitted)
Cfg_Latch	BOOL		Latching alarm: active until Cmd_Reset after RTN (ISA-18.2 state E)
Cfg_NoAck	BOOL		No acknowledge required — use only for alarms rationalised as such
Cfg_OnDly	TIME	T#0S	On-delay (ISA-18.2 chattering control)

Table I.9 continued

Member	Type	Init	Description
Cfg_OffDly	TIME	T#0S	Off-delay (ISA-18.2 chattering control)
Cfg_Priority	INT	3	Rationalised priority: 1=Emergency 2=High 3=Medium 4=Low
Cfg_Class	INT	0	Alarm class (alarm philosophy): 0=General 1=Safety 2=Environmental 3=Quality 4=Asset
Cfg_MaxShelf	TIME	T#8H	Maximum shelve duration (shelving must be time-limited)
Cmd_Ack	BOOL		Operator acknowledge (one-shot)
Cmd_Reset	BOOL		Reset latched alarm (one-shot)
Cmd_Shelve	BOOL		Operator shelve (one-shot)
Cmd_Unshelve	BOOL		Operator unshelve (one-shot)
Cmd_RemoveSvc	BOOL		Remove from service for maintenance (one-shot, permit required)
Cmd_ReturnSvc	BOOL		Return to service (one-shot)
Sts_Active	BOOL		Alarm active (after delays and latch)
Sts_Unack	BOOL		Alarm unacknowledged
Sts_Shelved	BOOL		Shelved by operator (ISA-18.2 state F)
Sts_Suppressed	BOOL		Suppressed by design (ISA-18.2 state G)
Sts_OOS	BOOL		Out of service (ISA-18.2 state H)
Sts_State	INT		0=A Normal 1=B Unack 2=C Acked 3=D RTN-Unack 4=E Latched 5=F Shelved 6=G Suppressed-by-design 7=H Out-of-service
Sts_ShelfLeft	TIME		Shelve time remaining
Val_Count	UDINT		Activation counter (alarm performance KPIs)
Val_LastActive	DT		Date/time of last activation
Val_LastRTN	DT		Date/time of last return to normal

I.7.2 T_AlmLim

Analog limit alarm: limit + deadband + ISA-18.2 / IEC 62682 alarm.

ISA-88 role: Support (alarm object) **MTP:** e.g. VAHLim / VAHEn / VAHAct **Standards:** ISA-18.2 / IEC 62682, ISA-101.01

Table I.10: T_AlmLim Members

Member	Type	Init	Description
Cfg_Lim	REAL	0.0	Alarm limit (EU) — from rationalisation / master alarm database
Cfg_DB	REAL	0.0	Deadband for return to normal (EU, ISA-18.2 chattering control)
Inp_Cond	BOOL		Raw alarm condition (written by owner logic)
Inp_Suppress	BOOL		Suppressed-by-design: state-based suppression by logic (ISA-18.2 state G)
Cfg_En	BOOL		Alarm enabled — set TRUE once Cfg_Lim is rationalised
Cfg_Latch	BOOL		Latching alarm: active until Cmd_Reset after RTN (ISA-18.2 state E)
Cfg_NoAck	BOOL		No acknowledge required — use only for alarms rationalised as such

Table I.10 continued

Member	Type	Init	Description
Cfg_OnDly	TIME	T#0S	On-delay (ISA-18.2 chattering control)
Cfg_OffDly	TIME	T#0S	Off-delay (ISA-18.2 chattering control)
Cfg_Priority	INT	3	Rationalised priority: 1=Emergency 2=High 3=Medium 4=Low
Cfg_Class	INT	0	Alarm class (alarm philosophy): 0=General 1=Safety 2=Environmental 3=Quality 4=Asset
Cfg_MaxShelf	TIME	T#8H	Maximum shelve duration (shelving must be time-limited)
Cmd_Ack	BOOL		Operator acknowledge (one-shot)
Cmd_Reset	BOOL		Reset latched alarm (one-shot)
Cmd_Shelve	BOOL		Operator shelve (one-shot)
Cmd_Unshelve	BOOL		Operator unshelve (one-shot)
Cmd_RemoveSvc	BOOL		Remove from service for maintenance (one-shot, permit required)
Cmd_ReturnSvc	BOOL		Return to service (one-shot)
Sts_Active	BOOL		Alarm active (after delays and latch)
Sts_Unack	BOOL		Alarm unacknowledged
Sts_Shelved	BOOL		Shelved by operator (ISA-18.2 state F)
Sts_Suppressed	BOOL		Suppressed by design (ISA-18.2 state G)
Sts_OOS	BOOL		Out of service (ISA-18.2 state H)
Sts_State	INT		0=A Normal 1=B Unack 2=C Acked 3=D RTN-Unack 4=E Latched 5=F Shelved 6=G Suppressed-by-design 7=H Out-of-service
Sts_ShelfLeft	TIME		Shelve time remaining
Val_Count	UDINT		Activation counter (alarm performance KPIs)
Val_LastActive	DT		Date/time of last activation
Val_LastRTN	DT		Date/time of last return to normal

I.7.3 T_Mode

Control-module mode and ISA-88 ownership (arbitration); maps to MTP OpMode.

ISA-88 role: Support (mode + ownership) **MTP:** OpMode (Offline/Operator/Automatic) + SrcMode **Standards:** ISA-88 / IEC 61512-1, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.11: T_Mode Members

Member	Type	Init	Description
Cmd_Auto	BOOL		Operator request Automatic (one-shot)
Cmd_Man	BOOL		Operator request Manual (one-shot)
Cmd_Maint	BOOL		Request Maintenance (one-shot)
Cmd_OOS	BOOL		Request Out-of-Service (one-shot)
Prog_Auto	BOOL		Owner request Automatic
Prog_Man	BOOL		Owner request Manual
Prog_Lock	BOOL		Owner locks current mode (operator cannot change)
Prog_Acquire	BOOL		ISA-88 arbitration: owner (phase / EM) acquires this module
Prog_Release	BOOL		ISA-88 arbitration: owner releases this module

Table I.11 continued

Member	Type	Init	Description
Inp_ReqOwnerID	DINT		ID of the phase / equipment module requesting ownership
Cmd_ForceRelease	BOOL		Supervisor forces release of ownership (one-shot)
Cfg_PowerUpMode	INT	2	Mode at cold start: 0=OOS 1=Maint 2=Man 3=Auto
Cfg_NoMaint	BOOL		Maintenance mode not allowed
Sts_Auto	BOOL		In Automatic (commanded by owner) — MTP Automatic
Sts_Man	BOOL		In Manual (commanded by operator) — MTP Operator
Sts_Maint	BOOL		In Maintenance
Sts_OOS	BOOL		Out of service — MTP Offline
Sts_Mode	INT		0=OOS 1=Maint 2=Man 3=Auto (MTP: 0=Offline 2=Operator 3=Automatic)
Sts_Owned	BOOL		Acquired by an owner
Val_OwnerID	DINT		ID of current owner, 0 = free

I.7.4 T_RunTime

Run-hours, starts and maintenance counters.

ISA-88 role: Support (asset data) **MTP:** - **Standards:** NAMUR NE107 (feeds Maintenance required)

Table I.12: T_RunTime Members

Member	Type	Init	Description
Inp_Running	BOOL		Equipment running (from owner logic)
Cfg_MaintHrs	REAL	0.0	Maintenance interval in run hours, 0 = off
Cmd_ResetMaint	BOOL		Reset hours-since-maintenance (one-shot)
Cmd_ResetAll	BOOL		Reset all counters (one-shot, supervisor)
Val_RunSecs	UDINT		Total run time (s) — master counter
Val_RunHrs	REAL		Total run hours (display)
Val_Starts	UDINT		Total starts / cycles
Val_StartsLastHr	INT		Starts in the last 60 min (rolling)
Val_LastRun	TIME		Duration of last run
Val_HrsSinceMaint	REAL		Run hours since last maintenance reset
Sts_MaintDue	BOOL		Maintenance due

I.8 Core Instruments and Devices

I.8.1 T_AI

Analog input: scaling, filtering, NE43 range check, alarms, sim.

ISA-88 role: Control module **MTP:** AnaMon **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP), NAMUR NE43

Table I.13: T_AI Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_EU	STRING[12]		Engineering units text
Inp_Raw	INT		Raw counts from analog channel
Inp_ChFault	BOOL		Channel / module diagnostic fault
Cfg_RawMin	INT	0	Raw counts at 0% (M340/M580 4-20mA default 0)
Cfg_RawMax	INT	10000	Raw counts at 100% (M340/M580 4-20mA default 10000)
Cfg_EUMin	REAL	0.0	PV at 0% (EU)
Cfg_EUMax	REAL	100.0	PV at 100% (EU)
Cfg_FiltTC	REAL	0.0	1st-order filter time constant (s), 0 = off
Cfg_SqrtExtract	BOOL		Square-root extraction (DP flow)
Cfg_LoCutoff	REAL	0.0	Low cutoff (% span): PV forced to EUMin below this, 0 = off
Cfg_BadLoPct	REAL	-2.5	Bad below this % span (NE43 3.6mA = -2.5%)
Cfg_BadHiPct	REAL	106.25	Bad above this % span (NE43 21mA = 106.25%)
Cfg_BadAction	INT	0	On bad PV: 0=Hold last good 1=Substitute 2=EUMin 3=EUMax
Cfg_SubstPV	REAL	0.0	Substitute PV used when Cfg_BadAction = 1
Sim_En	BOOL		Simulation enabled
Sim_PV	REAL	0.0	Simulated PV (EU)
Val_PV	REAL		Process value (EU) — use this in logic
Val_PVPct	REAL		Process value (% of span)
Val_ROC	REAL		Rate of change (EU/s, 1 s window) — input to Alm_ROC
Sts_Bad	BOOL		PV bad (out of range or channel fault)
Sts_URng	BOOL		Under range
Sts_ORng	BOOL		Over range
Sts_Sim	BOOL		Simulation active
Sts_Subst	BOOL		Substitute / held value in use
Sts_Quality	INT		0=Good 1=Uncertain (simulation, NE107 C or S) 2=Bad
Inp_NE107	INT	0	NE107 status reported by smart device (HART / fieldbus)
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Alm_HH	T_AlmLim		High-high alarm
Alm_H	T_AlmLim		High alarm
Alm_L	T_AlmLim		Low alarm
Alm_LL	T_AlmLim		Low-low alarm
Alm_ROC	T_AlmLim		Rate-of-change alarm (Cfg_Lim in EU/s)
Alm_Bad	T_Alarm		Bad PV alarm

I.8.2 T_DI

Digital input: invert, debounce, sim, optional state alarm.

ISA-88 role: Control module **MTP:** BinMon **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.14: T_DI Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Inp_Raw	BOOL		Raw input from channel
Inp_ChFault	BOOL		Channel / module diagnostic fault
Cfg_Invert	BOOL		Invert raw signal (fail-safe NC contacts)
Cfg_OnDly	TIME	T#0S	Debounce on-delay
Cfg_OffDly	TIME	T#0S	Debounce off-delay
Cfg_AlmMode	INT	0	0=No alarm 1=Alarm when Val TRUE 2=Alarm when Val FALSE
Cfg_TrueText	STRING[16]		HMI text for TRUE state, e.g. OPEN
Cfg_FalseText	STRING[16]		HMI text for FALSE state, e.g. CLOSED
Sim_En	BOOL		Simulation enabled
Sim_Val	BOOL		Simulated value
Val	BOOL		Conditioned value — use this in logic
Sts_Bad	BOOL		Channel fault
Sts_Sim	BOOL		Simulation active
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Val_Transitions	UDINT		Count of FALSE->TRUE transitions
Alm	T_Alarm		State alarm (see Cfg_AlmMode)
Alm_Bad	T_Alarm		Channel fault alarm

I.8.3 T_DValve

On/off valve: single or dual solenoid, limit switches, travel alarms.

ISA-88 role: Control module **MTP:** MonBinVlv **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.15: T_DValve Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Mode	T_Mode		Mode block
Inp_ZSO	BOOL		Open limit switch
Inp_ZSC	BOOL		Closed limit switch
Inp_Remote	BOOL		Local/Remote switch in REMOTE (if fitted)
Inp_IntlkOK	BOOL		Interlocks healthy (FALSE drives to interlock position)
Inp_PermOK	BOOL		Permissives healthy (TRUE = may leave safe position)
Inp_IOFault	BOOL		I/O channel or comms fault
Cfg_HasZSO	BOOL	TRUE	Open limit switch fitted
Cfg_HasZSC	BOOL	TRUE	Closed limit switch fitted
Cfg_FailOpen	BOOL		Fail position: TRUE = fail open, FALSE = fail closed
Cfg_IntlkOpen	BOOL		Interlock drives valve: TRUE = open, FALSE = closed
Cfg_DualCoil	BOOL		Dual solenoid (uses Out_Open and Out_Close)
Cfg_TravelTime	TIME	T#10S	Maximum travel time before failure alarm

Table I.15 continued

Member	Type	Init	Description
Cmd_Open	BOOL		Operator open (one-shot)
Cmd_Close	BOOL		Operator close (one-shot)
Cmd_Reset	BOOL		Fault reset (one-shot)
Prog_Open	BOOL		Program open request (Auto)
Prog_Close	BOOL		Program close request (Auto)
Out_Open	BOOL		Solenoid output: open (energise)
Out_Close	BOOL		Solenoid output: close (dual-coil only)
Sim_En	BOOL		Simulation: ignore field I/O, logic generates feedback
Sts_Opened	BOOL		Confirmed open
Sts_Closed	BOOL		Confirmed closed
Sts_Opening	BOOL		Travelling open
Sts_Closing	BOOL		Travelling closed
Sts_Avail	BOOL		Available to Auto/program control
Sts_Intlk	BOOL		Interlock not healthy
Sts_NotPerm	BOOL		Permissive not healthy
Sts_Local	BOOL		In local control
Sts_Fault	BOOL		Any fault present
Sts_Sim	BOOL		Simulation active
Sts_State	INT		0=Transit/unknown 1=Closed 2=Open 3=Opening 4=Closing 5=Fault
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Alm_FailOpen	T_Alarm		Failed to open within travel time
Alm_FailClose	T_Alarm		Failed to close within travel time
Alm_PosConflict	T_Alarm		Both limit switches made
Alm_Uncmd	T_Alarm		Left commanded position without a command
Alm_IntlkTrip	T_Alarm		Driven to safe position by interlock
Run	T_RunTime		Cycle counter (Val_Starts = operations)

I.8.4 T_CValve

Modulating control valve: limits, rate limit, positioner feedback.

ISA-88 role: Control module **MTP:** MonAnaVlv **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.16: T_CValve Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Mode	T_Mode		Mode block
Inp_PosFbk	REAL		Position feedback (%)
Inp_FbkBad	BOOL		Feedback signal bad
Inp_ZSC	BOOL		Closed limit switch (optional)
Inp_IntlkOK	BOOL		Interlocks healthy (FALSE drives to Cfg_IntlkPos)
Inp_IOFault	BOOL		I/O channel or comms fault

Table I.16 continued

Member	Type	Init	Description
Cfg_HasFbk	BOOL		Position feedback fitted
Cfg_FailPos	REAL	0.0	Fail position (%): 0 = fail closed, 100 = fail open
Cfg_IntlkPos	REAL	0.0	Position driven to on interlock (%)
Cfg_OutMin	REAL	0.0	Minimum output (%)
Cfg_OutMax	REAL	100.0	Maximum output (%)
Cfg_RateUp	REAL	0.0	Opening rate limit (%/s), 0 = none
Cfg_RateDn	REAL	0.0	Closing rate limit (%/s), 0 = none
Cfg_TightShut	REAL	0.0	Demand below this (%) forces 0 for tight shut-off
Cfg_ReverseAct	BOOL		Reverse-acting output (100% signal = closed)
Cfg_RawMin	INT	0	Analog output raw counts at 0%
Cfg_RawMax	INT	10000	Analog output raw counts at 100%
Cfg_DevLim	REAL	5.0	Position deviation alarm limit (%)
Cfg_DevTime	TIME	T#15S	Deviation must persist this long
Cmd_ManPos	REAL	0.0	Operator manual demand (%)
Cmd_Reset	BOOL		Fault reset (one-shot)
Prog_Pos	REAL	0.0	Program / PID demand (%) — Auto mode
Out_CV	REAL		Output demand (%) after limits and rate limit
Out_CVRaw	INT		Output in raw counts for AO channel
Sim_En	BOOL		Simulation: ignore field I/O, logic generates feedback
Val_Demand	REAL		Active demand before rate limit (%)
Val_Pos	REAL		Position (%) — feedback, or demand if no feedback
Val_Dev	REAL		Demand minus position (%)
Val_Travel	REAL		Accumulated travel (% of full stroke, summed) for diagnostics
Val_Reversals	UDINT		Direction reversal count
Sts_Closed	BOOL		Valve closed
Sts_Intlk	BOOL		Interlock active
Sts_RateLim	BOOL		Output is being rate limited
Sts_Fault	BOOL		Any fault present
Sts_Sim	BOOL		Simulation active
Inp_NE107	INT	0	NE107 status reported by smart device (HART / fieldbus)
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Alm_PosDev	T_Alarm		Position deviation
Alm_FbkBad	T_Alarm		Feedback bad
Alm_IntlkTrip	T_Alarm		Driven to interlock position

I.8.5 T_Motor

Direct-on-line / star-delta / soft-start motor.

ISA-88 role: Control module **MTP:** MonBinDrv **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.17: T_Motor Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Mode	T_Mode		Mode block
Inp_RunFbk	BOOL		Run feedback (contactor aux / drive running)
Inp_Remote	BOOL		Local/Remote switch in REMOTE
Inp_AvailOK	BOOL		MCC available: breaker closed, control power OK
Inp_EStopOK	BOOL		Field E-stop circuit healthy
Inp_IntlkOK	BOOL		Interlocks healthy (TRUE = may run; FALSE trips)
Inp_PermOK	BOOL		Permissives healthy (TRUE = may start)
Inp_IOFault	BOOL		I/O channel or comms fault
Inp_OL	BOOL		Overload / motor protection relay tripped
Inp_Current	REAL		Motor current (A), optional
Cfg_FbkTime	TIME	T#5S	Time allowed for run feedback after start/stop
Cfg_RestartDly	TIME	T#0S	Minimum stopped time before restart
Cfg_NoFbk	BOOL		No run feedback wired (feedback follows output)
Cfg_AutoRestart	BOOL		Auto mode restarts after power loss / interlock clears
Cfg_RatedAmps	REAL	0.0	Motor nameplate full-load current (A)
Cfg_MaxStartsHr	INT	0	Maximum starts per hour, 0 = no limit
Cmd_Start	BOOL		Operator start (one-shot)
Cmd_Stop	BOOL		Operator stop (one-shot)
Cmd_Reset	BOOL		Fault / trip reset (one-shot)
Cmd_Jog	BOOL		Maintenance jog (maintained, Maint mode only)
Prog_Start	BOOL		Program / sequence start request (Auto mode)
Prog_Stop	BOOL		Program / sequence stop request (Auto mode)
Out_Run	BOOL		Run output to contactor / starter
Sim_En	BOOL		Simulation: ignore field I/O, logic generates feedback
Sts_Running	BOOL		Running (feedback confirmed)
Sts_Stopped	BOOL		Stopped (feedback confirmed)
Sts_Starting	BOOL		Start commanded, waiting for feedback
Sts_Stopping	BOOL		Stop commanded, waiting for feedback
Sts_Avail	BOOL		Available to Auto/program control
Sts_Ready	BOOL		Ready to start now (all permissives, no trip)
Sts_Intlk	BOOL		Interlock not healthy
Sts_NotPerm	BOOL		Permissive not healthy
Sts_Local	BOOL		In local control
Sts_Tripped	BOOL		Tripped — reset required
Sts_Fault	BOOL		Any fault present
Sts_Sim	BOOL		Simulation active
Sts_State	INT		0=Stopped 1=Starting 2=Running 3=Stopping 4=Tripped 5=Not available
Sts_RestartLeft	TIME		Restart-delay time remaining
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Val_Current	REAL		Motor current (A)
Val_LoadPct	REAL		Load (% of Cfg_RatedAmps)
Alm_FailStart	T_Alarm		No run feedback within Cfg_FbkTime
Alm_FailStop	T_Alarm		Run feedback still present after stop

Table I.17 continued

Member	Type	Init	Description
Alm_Uncmd	T_Alarm		Running without a command
Alm_EStop	T_Alarm		E-stop activated
Alm_IntlkTrip	T_Alarm		Tripped by interlock
Alm_ExcessStarts	T_Alarm		Start blocked: Cfg_MaxStartsHr exceeded
Alm_OL	T_Alarm		Overload trip
Alm_HiCurrent	T_AlmLim		High current (A)
Run	T_RunTime		Run hours / starts

I.8.6 T_VFD

Variable speed drive (fieldbus or hardwired).

ISA-88 role: Control module **MTP:** MonAnaDrv **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.18: T_VFD Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Mode	T_Mode		Mode block
Inp_RunFbk	BOOL		Run feedback (contactor aux / drive running)
Inp_Remote	BOOL		Local/Remote switch in REMOTE
Inp_AvailOK	BOOL		MCC available: breaker closed, control power OK
Inp_EStopOK	BOOL		Field E-stop circuit healthy
Inp_IntlkOK	BOOL		Interlocks healthy (TRUE = may run; FALSE trips)
Inp_PermOK	BOOL		Permissives healthy (TRUE = may start)
Inp_IOFault	BOOL		I/O channel or comms fault
Inp_DrvReady	BOOL		Drive ready
Inp_DrvFault	BOOL		Drive faulted
Inp_DrvWarn	BOOL		Drive warning / alarm
Inp_FaultCode	UINT		Drive fault code
Inp_CommOK	BOOL	TRUE	Drive communications healthy
Inp_NE107	INT	0	NE107 status reported by smart device (HART / fieldbus)
Inp_SpdFbk	REAL		Speed feedback (Cfg_SpdEU)
Inp_Current	REAL		Motor current (A)
Inp_Torque	REAL		Motor torque (%)
Inp_Power	REAL		Motor power (kW)
Cfg_FbkTime	TIME	T#5S	Time allowed for run feedback after start/stop
Cfg_RestartDly	TIME	T#0S	Minimum stopped time before restart
Cfg_NoFbk	BOOL		No run feedback wired (feedback follows output)
Cfg_AutoRestart	BOOL		Auto mode restarts after power loss / interlock clears
Cfg_RatedAmps	REAL	0.0	Motor nameplate full-load current (A)
Cfg_MaxStartsHr	INT	0	Maximum starts per hour, 0 = no limit
Cfg_SpdEU	STRING[8]		Speed units text (%, Hz, rpm)
Cfg_SpdMin	REAL	0.0	Minimum speed reference
Cfg_SpdMax	REAL	100.0	Maximum speed reference

Table I.18 continued

Member	Type	Init	Description
Cfg_SpdDevLim	REAL	5.0	Speed deviation alarm limit
Cfg_SpdDevTime	TIME	T#20S	Speed deviation delay
Cfg_Reversible	BOOL		Reverse direction allowed
Cmd_Start	BOOL		Operator start (one-shot)
Cmd_Stop	BOOL		Operator stop (one-shot)
Cmd_Reset	BOOL		Fault / trip reset (one-shot)
Cmd_Jog	BOOL		Maintenance jog (maintained, Maint mode only)
Prog_Start	BOOL		Program / sequence start request (Auto mode)
Prog_Stop	BOOL		Program / sequence stop request (Auto mode)
Cmd_Spd	REAL	0.0	Operator manual speed reference
Cmd_Rev	BOOL		Operator direction: TRUE = reverse
Prog_Spd	REAL	0.0	Program / PID speed reference (Auto)
Prog_Rev	BOOL		Program direction: TRUE = reverse
Out_Run	BOOL		Run command to drive
Out_Rev	BOOL		Reverse command to drive
Out_SpdRef	REAL		Speed reference to drive (limited)
Out_FltReset	BOOL		Fault reset pulse to drive
Sim_En	BOOL		Simulation: ignore field I/O, logic generates feedback
Sts_Running	BOOL		Running (feedback confirmed)
Sts_Stopped	BOOL		Stopped (feedback confirmed)
Sts_Starting	BOOL		Start commanded, waiting for feedback
Sts_Stopping	BOOL		Stop commanded, waiting for feedback
Sts_Avail	BOOL		Available to Auto/program control
Sts_Ready	BOOL		Ready to start now (all permissives, no trip)
Sts_Intlk	BOOL		Interlock not healthy
Sts_NotPerm	BOOL		Permissive not healthy
Sts_Local	BOOL		In local control
Sts_Tripped	BOOL		Tripped — reset required
Sts_Fault	BOOL		Any fault present
Sts_Sim	BOOL		Simulation active
Sts_State	INT		0=Stopped 1=Starting 2=Running 3=Stopping 4=Tripped 5=Not available
Sts_RestartLeft	TIME		Restart-delay time remaining
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Sts_AtSpd	BOOL		At speed (feedback within deviation limit)
Sts_Rev	BOOL		Running in reverse
Sts_CommFail	BOOL		Communication failure
Val_SpdRef	REAL		Active speed reference
Val_Spd	REAL		Speed
Val_Current	REAL		Motor current (A)
Val_Torque	REAL		Torque (%)
Val_Power	REAL		Power (kW)
Val_Energy	LREAL		Energy (kWh), integrated from power
Alm_FailStart	T_Alarm		No run feedback within Cfg_FbkTime
Alm_FailStop	T_Alarm		Run feedback still present after stop
Alm_Uncmd	T_Alarm		Running without a command
Alm_EStop	T_Alarm		E-stop activated

Table I.18 continued

Member	Type	Init	Description
Alm_IntlkTrip	T_Alarm		Tripped by interlock
Alm_ExcessStarts	T_Alarm		Start blocked: Cfg_MaxStartsHr exceeded
Alm_DrvFault	T_Alarm		Drive fault
Alm_DrvWarn	T_Alarm		Drive warning
Alm_CommFail	T_Alarm		Drive comms failure
Alm_SpdDev	T_Alarm		Speed deviation
Run	T_RunTime		Run hours / starts

I.8.7 T_PID

PID loop wrapper (maps onto PIDFF EFB), Man/Auto/Cas, tracking.

ISA-88 role: Control module **MTP:** PIDCtrl **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.19: T_PID Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_PVEU	STRING[12]		PV engineering units text
Inp_PV	REAL		Process value (EU)
Inp_PVBad	BOOL		PV bad quality
Inp_CasSP	REAL		Cascade (remote) setpoint from master loop
Inp_FF	REAL		Feed-forward input (% CV)
Inp_TrackEn	BOOL		Output tracking enable
Inp_TrackVal	REAL		Output tracking value (%)
Inp_IntlkOK	BOOL		Interlocks healthy (FALSE forces Cfg_IntlkCV)
Cmd_Man	BOOL		Operator request Manual (one-shot)
Cmd_Auto	BOOL		Operator request Auto (one-shot)
Cmd_Cas	BOOL		Operator request Cascade (one-shot)
Cmd_SP	REAL	0.0	Operator setpoint (EU)
Cmd_CV	REAL	0.0	Operator manual output (%)
Prog_SP	REAL	0.0	Program setpoint (EU), used when Cfg_ProgSP
Cfg_ProgSP	BOOL		Setpoint owned by program instead of operator
Cfg_Kp	REAL	1.0	Proportional gain
Cfg_Ti	TIME	T#10S	Integral time, T#0S = no integral
Cfg_Td	TIME	T#0S	Derivative time
Cfg_DirectAct	BOOL		TRUE = direct acting (CV rises with PV); FALSE = reverse
Cfg_PVMin	REAL	0.0	PV range low (EU)
Cfg_PVMax	REAL	100.0	PV range high (EU)
Cfg_SPMin	REAL	0.0	Setpoint low limit (EU)
Cfg_SPMax	REAL	100.0	Setpoint high limit (EU)
Cfg_CVMin	REAL	0.0	Output low limit (%)
Cfg_CVMax	REAL	100.0	Output high limit (%)
Cfg_SPRate	REAL	0.0	Setpoint ramp rate (EU/s), 0 = step
Cfg_CVRate	REAL	0.0	Output rate limit (%/s), 0 = none

Table I.19 continued

Member	Type	Init	Description
Cfg_DB	REAL	0.0	Error deadband (EU)
Cfg_FFGain	REAL	0.0	Feed-forward gain
Cfg_PVTrack	BOOL		Setpoint tracks PV while in Manual (bumpless)
Cfg_IntlkCV	REAL	0.0	Output forced on interlock (%)
Cfg_BadPVAction	INT	1	On bad PV: 0=Hold CV 1=Go Manual 2=Go Cfg_IntlkCV
Out_CV	REAL		Controller output (%)
Val_SP	REAL		Active setpoint after ramp and limits (EU)
Val_SPTarget	REAL		Target setpoint before ramp (EU)
Val_PV	REAL		Process value (EU)
Val_Err	REAL		Error SP — PV (EU)
Sts_Man	BOOL		In Manual
Sts_Auto	BOOL		In Auto
Sts_Cas	BOOL		In Cascade
Sts_Track	BOOL		Output tracking
Sts_Intlk	BOOL		Interlock active
Sts_WindupHi	BOOL		Output at high limit
Sts_WindupLo	BOOL		Output at low limit
Sts_SPRamping	BOOL		Setpoint ramping
Sts_Mode	INT		0=Man 1=Auto 2=Cas 3=Track 4=Intlk
Alm_DevH	T_AlmLim		High deviation (PV — SP), EU
Alm_DevL	T_AlmLim		Low deviation (SP — PV), EU
Alm_PVBad	T_Alarm		PV bad

I.9 Rotating Equipment

I.9.1 T_PumpProt

Pump protection layer: dry run, low flow, deadhead, seal, bearing.

ISA-88 role: Control module **MTP:** - **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.20: T_PumpProt Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Inp_Running	BOOL		Pump running (from Motor / VFD Sts_Running)
Inp_SuctPress	REAL		Suction pressure (EU)
Inp_DischPress	REAL		Discharge pressure (EU)
Inp_Flow	REAL		Pump flow (EU)
Inp_DryRun	BOOL		Dry-run switch / low level active
Inp_SealLeak	BOOL		Seal leak / seal pot alarm
Inp_BrgTemp	REAL		Bearing temperature (degC)
Inp_Vib	REAL		Vibration (mm/s)
Cfg_MinFlow	REAL	0.0	Minimum continuous flow (EU)
Cfg_MinFlowDly	TIME	T#30S	Low-flow delay after start
Cfg_DeadheadDP	REAL	0.0	Differential pressure indicating deadhead (EU)

Table I.20 continued

Member	Type	Init	Description
Cfg_DryRunDly	TIME	T#5S	Dry-run delay
Cfg_TripDryRun	BOOL	TRUE	Trip on dry run
Cfg_TripLowFlow	BOOL		Trip on low flow
Cfg_TripDeadhead	BOOL	TRUE	Trip on deadhead
Cfg_TripSeal	BOOL		Trip on seal leak
Cmd_Reset	BOOL		Reset latched trips (one-shot)
Out_IntlkOK	BOOL		Protections healthy — wire to Motor/VFD Inp_IntlkOK
Val_DP	REAL		Differential pressure (EU)
Alm_DryRun	T_Alarm		Dry run
Alm_LowFlow	T_Alarm		Low flow (below minimum continuous flow)
Alm_Deadhead	T_Alarm		Deadhead (high DP, no flow)
Alm_SealLeak	T_Alarm		Seal leak
Alm_BrgTemp	T_AlmLim		High bearing temperature
Alm_Vib	T_AlmLim		High vibration
Alm_LoSuct	T_AlmLim		Low suction pressure

I.9.2 T_PumpDS

Duty/standby/lead-lag coordinator for up to 4 pumps.

ISA-88 role: Equipment module **MTP:** - **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.21: T_PumpDS Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Mode	T_Mode		Mode block (Auto = Prog_NumReq, Man = Cmd_NumReq)
Inp_Avail	ARRAY[1..4] OF BOOL		Pump n available (Motor Sts_Avail)
Inp_Running	ARRAY[1..4] OF BOOL		Pump n running
Inp_Fault	ARRAY[1..4] OF BOOL		Pump n faulted
Inp_RunHrs	ARRAY[1..4] OF REAL		Pump n run hours (for equalisation)
Cfg_NumPumps	INT	2	Number of pumps in group (1-4)
Cfg_Strategy	INT	1	Rotation: 0=Fixed lead 1=Rotate on stop 2=Equalise hours 3=Timed
Cfg_RotateHrs	REAL	168.0	Timed rotation interval (h)
Cfg_StageDly	TIME	T#10S	Delay between staging pumps on/off
Cfg_ChgOnFault	BOOL	TRUE	Automatic changeover to standby on fault
Cmd_NumReq	INT	0	Operator: number of pumps required (Manual)
Cmd_Lead	INT	1	Operator-selected lead pump (Fixed strategy)
Cmd_Rotate	BOOL		Force rotation now (one-shot)
Prog_NumReq	INT	0	Program: number of pumps required (Auto)

Table I.21 continued

Member	Type	Init	Description
Out_RunReq	ARRAY[1..4] OF BOOL		Run request to pump n (wire to Prog_Start)
Sts_Order	ARRAY[1..4] OF INT		Rank of pump n: 1=Lead 2=Lag1 ...
Sts_Lead	INT		Current lead pump
Sts_NumRunning	INT		Pumps running
Sts_NumAvail	INT		Pumps available
Val_HrsToRotate	REAL		Hours until next timed rotation
Alm_NoStandby	T_Alarm		No standby pump available
Alm_NoneAvail	T_Alarm		Demand present but no pump available
Sts_ChgOver	BOOL		Automatic changeover occurred (event, not an alarm — ISA-18.2)
Val_ChgOverCount	UDINT		Automatic changeover counter

I.9.3 T_Agitator

Agitator / mixer: VFD + submergence, seal and gearbox protection.

ISA-88 role: Equipment module **MTP:** MonAnaDrv (drive) **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.22: T_Agitator Members

Member	Type	Init	Description
Drv	T_VFD		Drive (fixed-speed agitators: ignore speed members)
Inp_Level	REAL		Vessel level (EU)
Inp_SealOK	BOOL		Seal / seal support system healthy
Inp_GbxTemp	REAL		Gearbox oil temperature (degC)
Inp_Vib	REAL		Vibration (mm/s)
Cfg_MinLevel	REAL	0.0	Minimum level to run (impeller submergence)
Cfg_LvlHyst	REAL	0.0	Level hysteresis (EU)
Cfg_TripLowLvl	BOOL	TRUE	Trip running agitator on low level
Out_IntlkOK	BOOL		Wire to Drv.Inp_IntlkOK
Out_PermOK	BOOL		Wire to Drv.Inp_PermOK
Sts_LvlOK	BOOL		Level above minimum
Alm_LowLvlRun	T_Alarm		Running / start requested below minimum level
Alm_Seal	T_Alarm		Seal fault
Alm_GbxTemp	T_AlmLim		High gearbox temperature
Alm_Vib	T_AlmLim		High vibration

I.9.4 T_Compressor

Compressor / blower: load-unload, anti-cycle, lube and thermal protection.

ISA-88 role: Equipment module **MTP:** MonAnaDrv (drive) **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.23: T_Compressor Members

Member	Type	Init	Description
Drv	T_VFD		Drive (fixed-speed: ignore speed members)
Inp_SuctPress	REAL		Suction pressure (EU)
Inp_DischPress	REAL		Discharge / header pressure (EU)
Inp_DischTemp	REAL		Discharge temperature (degC)
Inp_OilPress	REAL		Lube oil pressure (EU)
Inp_OilTemp	REAL		Lube oil temperature (degC)
Inp_Vib	REAL		Vibration (mm/s)
Inp_LoadFbk	BOOL		Loaded feedback (inlet valve open)
Cfg_LoadPress	REAL	0.0	Load (cut-in) pressure
Cfg_UnloadPress	REAL	0.0	Unload (cut-out) pressure
Cfg_MinRun	TIME	T#5M	Minimum run time
Cfg_MinStop	TIME	T#2M	Minimum stop time (anti-cycle)
Cfg_IdleTime	TIME	T#10M	Unloaded run time before auto stop
Cfg_OilBypass	TIME	T#10S	Low oil pressure bypass after start
Out_Load	BOOL		Load solenoid / inlet valve
Out_BlowOff	BOOL		Blow-off / recycle valve open
Out_IntlkOK	BOOL		Wire to Drv.Inp_IntlkOK
Sts_Loaded	BOOL		Loaded
Sts_Idle	BOOL		Running unloaded
Sts_AntiCycle	BOOL		Waiting on minimum stop time
Sts_State	INT		0=Off 1=Starting 2=Unloaded 3=Loaded 4=Idle stop 5=Anti-cycle 6=Tripped
Val_TimeToStart	TIME		Anti-cycle time remaining
Val_PressRatio	REAL		Compression ratio (absolute)
Alm_HiDischPress	T_AlmLim		High discharge pressure
Alm_HiDischTemp	T_AlmLim		High discharge temperature
Alm_LoOilPress	T_AlmLim		Low lube oil pressure
Alm_HiOilTemp	T_AlmLim		High lube oil temperature
Alm_Vib	T_AlmLim		High vibration
Alm_LoSuct	T_AlmLim		Low suction pressure

I.9.5 T_Conveyor

Belt / screw conveyor: start warning, zero-speed, pull-cord, alignment.

ISA-88 role: Equipment module **MTP:** MonBinDrv (drive) **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.24: T_Conveyor Members

Member	Type	Init	Description
Mtr	T_Motor		Drive motor
Inp_MotionOK	BOOL		Zero-speed switch: motion detected
Inp_PullCordOK	BOOL		Pull-cord / lanyard switches healthy
Inp_AlignLOK	BOOL		Belt alignment left healthy
Inp_AlignROK	BOOL		Belt alignment right healthy
Inp_ChuteOK	BOOL		Chute not blocked

Table I.24 continued

Member	Type	Init	Description
Inp_BeltRipOK	BOOL		Belt rip detector healthy
Inp_DownstrRun	BOOL		Downstream conveyor running
Inp_Rate	REAL		Belt scale rate (t/h), optional
Cfg_UseDownstr	BOOL		Require downstream running (cascade start/trip)
Cfg_WarnTime	TIME	T#10S	Pre-start warning horn time
Cfg_MotionDly	TIME	T#5S	Time after start to detect motion
Cfg_PurgeTime	TIME	T#0S	Run-out time before stopping (empty belt)
Out_Horn	BOOL		Start warning horn
Out_Beacon	BOOL		Start warning beacon
Out_IntlkOK	BOOL		Wire to Mtr.Inp_IntlkOK
Out_PermOK	BOOL		Wire to Mtr.Inp_PermOK
Out_StartReq	BOOL		Start request to Mtr after warning completes
Sts_Warning	BOOL		Pre-start warning in progress
Sts_Purging	BOOL		Running out before stop
Sts_State	INT		0=Stopped 1=Warning 2=Starting 3=Running 4=Purging 5=Tripped
Val_Total	LREAL		Total conveyed (t)
Alm_ZeroSpd	T_Alarm		Zero-speed / belt slip
Alm_PullCord	T_Alarm		Pull-cord operated
Alm_Misalign	T_Alarm		Belt misalignment
Alm_Chute	T_Alarm		Chute blocked
Alm_BeltRip	T_Alarm		Belt rip
Alm_DownstrStop	T_Alarm		Tripped by downstream conveyor

I.9.6 T_DiaComp

Diaphragm compressor unit: one motor, two diaphragm stages, leak detection, hydraulic oil, purge.

ISA-88 role: Equipment module **MTP:** MonBinDrv (drive) **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01

Built for the PDC/Fluitron-style two-stage diaphragm compressor (Section 10.9.1 of the Controls Standard): one motor turns the crankshaft for both stages, so the unit owns a single nested `T_Motor` and drives it through `Out_StartReq`, `Out_StopReq`, `Out_IntlkOK`, and `Out_PermOK`. `Sts_FirstOut` records the first trip cause for the HMI. Trip setpoints and shutdown actions come from the vendor documentation and the hazard review, not from the initial values below.

Table I.25: T_DiaComp Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Mode	T_Mode		Unit mode block (Auto = Prog_ commands, Man = operator)
Mtr	T_Motor		Drive motor (one motor drives both stages through the crankshaft)
Cfg_PressEU	STRING[8]		Pressure units text, e.g. barg, psig
Inp_SuctPress	REAL		Stage 1 suction pressure (Cfg_PressEU)
Inp_InterPress	REAL		Interstage pressure: stage 1 discharge / stage 2 suction

Table I.25 continued

Member	Type	Init	Description
Inp_DischPress	REAL		Stage 2 (final) discharge pressure
Inp_S1DischTemp	REAL		Stage 1 discharge gas temperature (degC)
Inp_S2DischTemp	REAL		Stage 2 discharge gas temperature (degC)
Inp_S1OilPress	REAL		Stage 1 head hydraulic oil pressure (Cfg_PressEU)
Inp_S2OilPress	REAL		Stage 2 head hydraulic oil pressure (Cfg_PressEU)
Inp_CaseOilPress	REAL		Crankcase lube oil pressure (Cfg_PressEU)
Inp_CaseOilTemp	REAL		Crankcase lube oil temperature (degC)
Inp_CWTemp	REAL		Cooling water return temperature (degC)
Inp_S1LeakOK	BOOL		Stage 1 diaphragm leak detection healthy (no leak between diaphragms)
Inp_S2LeakOK	BOOL		Stage 2 diaphragm leak detection healthy (no leak between diaphragms)
Inp_CWFlowOK	BOOL		Cooling water flow switch healthy
Inp_GasDetOK	BOOL		Area gas detection healthy (from T_GasDet / ESD)
Inp_ExtIntlkOK	BOOL		External interlocks healthy (ESD, upstream / downstream)
Inp_SuctVlvOpen	BOOL		Suction isolation valve confirmed open
Inp_DischVlvOpen	BOOL		Discharge isolation valve confirmed open
Cfg_CtrlMode	INT	0	0=Start/stop by command only 1=Auto start/stop on Inp_DischPress 2=Run continuous, recycle on Inp_DischPress
Cfg_LoadPress	REAL	0.0	Load / start (cut-in) discharge pressure
Cfg_UnloadPress	REAL	0.0	Unload / stop (cut-out) discharge pressure
Cfg_UnloadedStart	TIME	T#10S	Run on recycle after start before loading
Cfg_OilBypass	TIME	T#10S	Low oil pressure trips bypassed after start
Cfg_MinRun	TIME	T#5M	Minimum run time
Cfg_MinStop	TIME	T#2M	Minimum stop time (anti-cycle)
Cfg_IdleTime	TIME	T#10M	Unloaded run time before auto stop (Cfg_CtrlMode 1)
Cfg_LeakDly	TIME	T#2S	Diaphragm leak confirmation delay
Cfg_TripOnLeak	BOOL	TRUE	Trip on diaphragm leak (FALSE = alarm only)
Cfg_PurgeOnLeak	BOOL		Start the purge automatically after a leak trip
Cfg_PurgeTime	TIME	T#5M	Purge duration
Cfg_DiaHrs	REAL	0.0	Diaphragm replacement interval (run hours), 0 = off
Cfg_PAtm	REAL	0.0	Added to gauge pressures for compression ratio; 0 if transmitters read absolute
Cmd_Start	BOOL		Operator start (one-shot)
Cmd_Stop	BOOL		Operator stop (one-shot)
Cmd_Reset	BOOL		Reset unit trips (one-shot)
Cmd_Load	BOOL		Operator load: close recycle (one-shot, Manual)
Cmd_Unload	BOOL		Operator unload: open recycle (one-shot, Manual)
Cmd_Purge	BOOL		Start purge (one-shot, unit stopped)
Cmd_PurgeStop	BOOL		Abort purge (one-shot)
Cmd_ResetDiaS1	BOOL		Stage 1 diaphragms replaced: reset hours (one-shot)
Cmd_ResetDiaS2	BOOL		Stage 2 diaphragms replaced: reset hours (one-shot)
Prog_Start	BOOL		Program / sequence start request (Auto)
Prog_Stop	BOOL		Program / sequence stop request (Auto)
Out_StartReq	BOOL		Start request — wire to Mtr.Prog_Start
Out_StopReq	BOOL		Stop request — wire to Mtr.Prog_Stop
Out_IntlkOK	BOOL		Unit trips healthy — wire to Mtr.Inp_IntlkOK

Table I.25 continued

Member	Type	Init	Description
Out_PermOK	BOOL		Start permissives healthy — wire to Mtr.Inp_PermOK
Out_Recycle	BOOL		Recycle / bypass valve open (unloaded start, unload)
Out_SuctVlv	BOOL		Open suction isolation valve
Out_DischVlv	BOOL		Open discharge isolation valve
Out_CWVlv	BOOL		Open cooling water valve
Out_PurgeVlv	BOOL		Open purge gas valve
Sim_En	BOOL		Simulation: ignore field I/O, logic generates feedback
Sts_State	INT		0=Stopped 1=Pre-start 2=Starting unloaded 3=Loaded 4=Unloaded (recycle) 5=Stopping 6=Anti-cycle 7=Purging 8=Tripped 9=Not available
Sts_Loaded	BOOL		Running loaded
Sts_Unloaded	BOOL		Running on recycle
Sts_AntiCycle	BOOL		Waiting on minimum stop time
Sts_Purging	BOOL		Purge in progress
Sts_Tripped	BOOL		Unit tripped — reset required
Sts_ResetReq	BOOL		Trip causes cleared, reset required
Sts_FirstOut	INT		First trip cause: 0=None 1=S1 leak 2=S2 leak 3=Case oil 4=S1 head oil 5=S2 head oil 6=S1 disch temp 7=S2 disch temp 8=Interstage press 9=Disch press 10=Suction press 11=Cooling water 12=Gas detection 13=External interlock 14=Motor
Sts_DiaDueS1	BOOL		Stage 1 diaphragm replacement due
Sts_DiaDueS2	BOOL		Stage 2 diaphragm replacement due
Val_TimeToStart	TIME		Anti-cycle time remaining
Val_PurgeLeft	TIME		Purge time remaining
Val_S1Ratio	REAL		Stage 1 compression ratio (absolute)
Val_S2Ratio	REAL		Stage 2 compression ratio (absolute)
Val_S1DiaHrs	REAL		Stage 1 run hours since diaphragm change
Val_S2DiaHrs	REAL		Stage 2 run hours since diaphragm change
Alm_S1Leak	T_Alarm		Stage 1 diaphragm leak
Alm_S2Leak	T_Alarm		Stage 2 diaphragm leak
Alm_LoCaseOil	T_AlmLim		Low crankcase oil pressure
Alm_HiCaseOilT	T_AlmLim		High crankcase oil temperature
Alm_LoS1Oil	T_AlmLim		Low stage 1 head hydraulic oil pressure
Alm_LoS2Oil	T_AlmLim		Low stage 2 head hydraulic oil pressure
Alm_HiS1DischT	T_AlmLim		High stage 1 discharge temperature
Alm_HiS2DischT	T_AlmLim		High stage 2 discharge temperature
Alm_HiInterPress	T_AlmLim		High interstage pressure
Alm_HiDischPress	T_AlmLim		High final discharge pressure
Alm_LoSuct	T_AlmLim		Low suction pressure
Alm_HiCWTemp	T_AlmLim		High cooling water return temperature
Alm_CWFail	T_Alarm		Cooling water flow failure
Alm_GasDet	T_Alarm		Unit tripped by area gas detection
Alm_ExtIntlk	T_Alarm		Unit tripped by external interlock
Alm_DiaDue	T_Alarm		Diaphragm replacement due (either stage)

I.10 Process Equipment

I.10.1 T_Tank

Tank / vessel: level-to-volume, inventory, fill/drain permissives, alarms.

ISA-88 role: Equipment module / Unit vessel **MTP:** - **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.26: T_Tank Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_Product	STRING[24]		Stored product
Cfg_LvlEU	STRING[8]		Level units text
Cfg_VolEU	STRING[8]		Volume units text
Inp_Level	REAL		Level (Cfg_LvlEU) from AI Val_PV
Inp_LevelBad	BOOL		Level signal bad
Inp_LSHH	BOOL		High-high level switch active
Inp_LSLL	BOOL		Low-low level switch active
Inp_Temp	REAL		Temperature (degC)
Inp_Press	REAL		Pressure (EU)
Cfg_MaxLevel	REAL	100.0	Level at 100% full
Cfg_Capacity	REAL	100.0	Volume at 100% full
Cfg_Shape	INT	0	0=Vertical / linear 1=Horizontal cylinder 2=Strapping table
Cfg_Strap	ARRAY[0..10] OF REAL		Strapping table: volume at 0,10..100% level
Cfg_SG	REAL	1.0	Specific gravity (for mass)
Cfg_FillStop	REAL	90.0	Fill permissive removed above this level
Cfg_DrainStop	REAL	5.0	Drain permissive removed below this level
Cfg_RateWindow	TIME	T#1M	Window for net-flow (rate of change) calc
Prog_EqState	INT		ISA-88 equipment state request from unit / phase logic
Out_FillPerm	BOOL		Filling permitted (wire to inlet pump / valve)
Out_DrainPerm	BOOL		Draining permitted (wire to outlet pump / valve)
Val_Volume	REAL		Volume (Cfg_VolEU)
Val_PctFull	REAL		Volume (% of capacity)
Val_Ullage	REAL		Remaining capacity (Cfg_VolEU)
Val_Mass	REAL		Mass (t, from volume x SG)
Val_NetFlow	REAL		Net in/out flow (Cfg_VolEU per min, + = filling)
Val_MinToFull	REAL		Minutes to full at current rate
Val_MinToEmpty	REAL		Minutes to empty at current rate
Sts_Filling	BOOL		Level rising
Sts_Draining	BOOL		Level falling
Sts_EqState	INT		ISA-88 equipment state: 0=Available-clean 1=In use 2=Dirty 3=Cleaning (CIP) 4=Sterilising (SIP) 5=Maintenance 6=Out of service
Alm_HH	T_AlmLim		High-high level
Alm_H	T_AlmLim		High level
Alm_L	T_AlmLim		Low level
Alm_LL	T_AlmLim		Low-low level
Alm_LSHH	T_Alarm		High-high level switch
Alm_LSLL	T_Alarm		Low-low level switch

Table I.26 continued

Member	Type	Init	Description
Alm_LvlDisc	T_Alarm		Analog level disagrees with level switch
Alm_HiTemp	T_AlmLim		High temperature
Alm_HiPress	T_AlmLim		High pressure

I.10.2 T_Totalizer

Flow totalizer + batch delivery: analog or pulse, day/hour totals.

ISA-88 role: Control module **MTP:** - **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.27: T_Totalizer Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_RateEU	STRING[12]		Rate units text, e.g. m3/h
Cfg_TotEU	STRING[12]		Total units text, e.g. m3
Inp_Rate	REAL		Flow rate (Cfg_RateEU)
Inp_RateBad	BOOL		Rate signal bad
Inp_Pulses	UDINT		Raw pulse counter (pulse mode)
Cfg_UsePulses	BOOL		Totalise from pulses instead of rate
Cfg_PulseWt	REAL	1.0	Total units per pulse
Cfg_TimeBase	REAL	3600.0	Seconds per rate time unit: 1=/s 60=/min 3600=/h
Cfg_Conv	REAL	1.0	Conversion rate units -> total units
Cfg_LoCutoff	REAL	0.0	Rates below this are not totalised
Cfg_BadAction	INT	0	On bad rate: 0=Stop totalising 1=Use last good rate
Cfg_DayStartHr	INT	0	Hour the production day starts (0-23)
Cfg_BatchSP	REAL	0.0	Batch target quantity
Cfg_PreAct	REAL	0.0	Pre-act (in-flight) quantity
Cfg_AutoPreAct	BOOL		Learn pre-act from overrun
Cfg_SlowPct	REAL	90.0	Dribble / slow-fill at this % of batch
Cmd_Start	BOOL		Operator batch start (one-shot)
Cmd_Hold	BOOL		Operator batch hold (one-shot)
Cmd_Reset	BOOL		Operator batch reset (one-shot)
Prog_Start	BOOL		Program batch start
Prog_Hold	BOOL		Program batch hold
Prog_Reset	BOOL		Program batch reset
Out_Slow	BOOL		Slow-fill / dribble
Out_BatchDone	BOOL		Batch target reached (close valve)
Val_Batch	LREAL		Batch total
Val_Remaining	REAL		Batch remaining
Val_Overrun	REAL		Last batch overrun
Val_GrandTotal	LREAL		Grand total (non-resettable)
Val_ThisHr	LREAL		Total this hour
Val_LastHr	LREAL		Total last hour
Val_Today	LREAL		Total this production day
Val_Yday	LREAL		Total previous production day

Table I.27 continued

Member	Type	Init	Description
Sts_Running	BOOL		Batch running
Sts_Hold	BOOL		Batch held
Sts_Done	BOOL		Batch complete
Alm_HiRate	T_AlmLim		High flow rate
Alm_LoRate	T_AlmLim		Low flow rate
Alm_Overrun	T_AlmLim		Batch overrun
Alm_RateBad	T_Alarm		Rate signal bad while totalising

I.10.3 T_HeatEx

Heat exchanger performance monitor: duty, LMTD, UA, fouling, DP.

ISA-88 role: Control module (monitoring) **MTP:** - **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.28: T_HeatEx Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Inp_HotInT	REAL		Hot side inlet temperature (degC)
Inp_HotOutT	REAL		Hot side outlet temperature (degC)
Inp_ColdInT	REAL		Cold side inlet temperature (degC)
Inp_ColdOutT	REAL		Cold side outlet temperature (degC)
Inp_HotFlow	REAL		Hot side mass flow (kg/s)
Inp_ColdFlow	REAL		Cold side mass flow (kg/s)
Inp_HotDP	REAL		Hot side differential pressure (kPa)
Inp_ColdDP	REAL		Cold side differential pressure (kPa)
Cfg_HotCp	REAL	4.186	Hot side specific heat (kJ/kg.K)
Cfg_ColdCp	REAL	4.186	Cold side specific heat (kJ/kg.K)
Cfg_CounterFlow	BOOL	TRUE	Counter-current (FALSE = co-current) for LMTD
Cfg_DesignUA	REAL	0.0	Clean design UA (kW/K)
Cfg_MinFlow	REAL	0.0	Below this flow the calculations are frozen (kg/s)
Cfg_CalcFilt	REAL	60.0	Filter time constant for calculated values (s)
Val_DutyHot	REAL		Heat released by hot side (kW)
Val_DutyCold	REAL		Heat absorbed by cold side (kW)
Val_Duty	REAL		Average duty (kW)
Val_HeatBal	REAL		Heat balance error (%)
Val_LMTD	REAL		Log mean temperature difference (K)
Val_UA	REAL		Actual UA (kW/K)
Val_Clean	REAL		Cleanliness: UA as % of design
Val_Approach	REAL		Approach temperature (K)
Val_Effect	REAL		Thermal effectiveness (0-1)
Sts_CalcValid	BOOL		Inputs valid, calculations live
Alm_Fouling	T_AlmLim		Low cleanliness (Val_Clean below limit)
Alm_HotDP	T_AlmLim		High hot side DP
Alm_ColdDP	T_AlmLim		High cold side DP
Alm_HeatBal	T_AlmLim		Heat balance error (possible instrument fault)

Table I.28 continued

Member	Type	Init	Description
Alm_HiOutT	T_AlmLim		High process outlet temperature

I.10.4 T_DosePump

Chemical dosing / metering pump: manual, flow-paced, trim, batch.

ISA-88 role: Equipment module **MTP:** MonAnaDrv (drive) **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.29: T_DosePump Members

Member	Type	Init	Description
Mtr	T_Motor		Pump motor / enable
Cfg_RateEU	STRING[8]		Dose rate units text, e.g. L/h
Cfg_DoseMode	INT	0	0=Manual rate 1=Flow-paced 2=Flow-paced + trim 3=Batch volume
Cfg_MaxRate	REAL	0.0	Pump rate at 100% output (L/h)
Cfg_CalFactor	REAL	1.0	Calibration factor from drawdown test
Cfg_DoseConc	REAL	0.0	Target dose in process (mg/L)
Cfg_ChemConc	REAL	100.0	Chemical strength (% w/w)
Cfg_ChemSG	REAL	1.0	Chemical specific gravity
Cfg_TankCap	REAL	0.0	Chemical day tank capacity (L)
Cfg_NoFlowDly	TIME	T#30S	Flow verification delay
Inp_PaceFlow	REAL		Process flow for pacing (m3/h)
Inp_TrimPct	REAL		Trim from residual PID (%, -100..+100)
Inp_FlowOK	BOOL		Flow verification switch: dosing detected
Inp_ChemLvl	REAL		Chemical day tank level (%)
Inp_Leak	BOOL		Leak detector / bund alarm
Cmd_Rate	REAL	0.0	Operator dose rate (L/h)
Prog_Rate	REAL	0.0	Program dose rate (L/h)
Cmd_CalDone	BOOL		Calibration complete, stamp date (one-shot)
Out_Rate	REAL		Speed / stroke output to pump (%)
Out_IntlkOK	BOOL		Wire to Mtr.Inp_IntlkOK
Val_RateSP	REAL		Active dose rate (L/h)
Val_Total	LREAL		Total dosed (L)
Val_Today	REAL		Dosed today (L)
Val_DaysLeft	REAL		Days of chemical remaining at current usage
Val_LastCal	DT		Date of last calibration
Sts_RateLimited	BOOL		Demand exceeds pump capacity
Alm_NoFlow	T_Alarm		No flow detected while running
Alm_LoChem	T_AlmLim		Low chemical level (%)
Alm_LoLoChem	T_AlmLim		Low-low chemical level (%) — stops pump
Alm_Leak	T_Alarm		Leak detected
Alm_RateLim	T_Alarm		Dose demand exceeds pump capacity

I.11 Analyzers and Safety

I.11.1 T_Analyzer

Process analyzer (pH, ORP, conductivity, DO, turbidity, Cl2 ...).

ISA-88 role: Control module **MTP:** AnaMon **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.30: T_Analyzer Members

Member	Type	Init	Description
AI	T_AI		Measured value, scaling and alarms
Cfg_AnType	INT	1	1=pH 2=ORP 3=Conductivity 4=DO 5=Turbidity 6=Free Cl2 7=TOC 8=Other
Inp_Temp	REAL		Sample / compensation temperature (degC)
Inp_DevFault	BOOL		Transmitter diagnostic fault
Inp_DevMaint	BOOL		Transmitter in maintenance / calibration
Inp_DevWarn	BOOL		Transmitter warning (e.g. sensor ageing)
Inp_NE107	INT	0	NE107 status reported by smart device (HART / field-bus)
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Inp_SampleOK	BOOL		Sample flow healthy (flow switch)
Inp_Slope	REAL		Sensor slope from transmitter (e.g. pH mV/pH)
Cfg_HoldMax	TIME	T#1H	Maximum hold time before alarm
Cfg_CalDays	INT	30	Calibration interval (days)
Cfg_FrozenTime	TIME	T#30M	PV unchanged for this long = frozen sensor
Cfg_FrozenBand	REAL	0.0	Band within which PV is considered unchanged
Cmd_Hold	BOOL		Hold PV and suppress alarms for calibration (one-shot)
Cmd_Release	BOOL		Release hold (one-shot)
Cmd_CalDone	BOOL		Calibration completed, stamp date (one-shot)
Val_PV	REAL		Value for control — held value while in hold
Val_HeldPV	REAL		Value captured at start of hold
Val_LastCal	DT		Date/time of last calibration
Val_DaysSinceCal	REAL		Days since last calibration
Sts_Hold	BOOL		In hold / calibration
Sts_CalDue	BOOL		Calibration due
Alm_DevFault	T_Alarm		Transmitter fault
Alm_NoSample	T_Alarm		No sample flow
Alm_Frozen	T_Alarm		Frozen / stuck reading
Alm_CalDue	T_Alarm		Calibration overdue
Alm_HoldLong	T_Alarm		Hold exceeded Cfg_HoldMax

I.11.2 T_GasDet

Fixed gas detector (4-20mA): latching A1/A2/A3, inhibit, fault.

ISA-88 role: Control module **MTP:** AnaMon **Standards:** ISA-88 / IEC 61512-1, ISA-5.1, ISA-18.2 / IEC 62682, ISA-101.01, NAMUR NE107, VDI/VDE/NAMUR 2658-4 (MTP), NAMUR NE43, ISA-84 / IEC 61511 (BPCS layer only)

Table I.31: T_GasDet Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_Gas	STRING[12]		Gas, e.g. CH4, H2S, CO, O2
Cfg_EU	STRING[8]		Units, e.g. %LEL, ppm, %vol
Inp_Raw	INT		Raw counts from analog channel
Inp_ChFault	BOOL		Channel / module diagnostic fault
Cfg_RawMin	INT	0	Raw counts at 4 mA
Cfg_RawMax	INT	10000	Raw counts at 20 mA
Cfg_RangeMax	REAL	100.0	Concentration at 20 mA
Cfg_O2Mode	BOOL		Oxygen depletion: A1/A2 act on falling concentration
Cfg_FaultmA	REAL	3.0	Signal below this (mA) = detector fault
Cfg_MaintLomA	REAL	0.0	Maintenance / inhibit signal band low (mA), vendor-specific
Cfg_MaintHimA	REAL	0.0	Maintenance / inhibit signal band high (mA), vendor-specific
Cfg_OverRngmA	REAL	20.5	Signal above this (mA) = over-range (treat as alarm)
Cfg_InhibitMax	TIME	T#2H	Maximum operator inhibit time
Cmd_Inhibit	BOOL		Operator inhibit for maintenance (one-shot)
Cmd_Uninhibit	BOOL		Remove inhibit (one-shot)
Cmd_Reset	BOOL		Reset latched alarms (one-shot)
Sim_En	BOOL		Simulation enabled
Sim_mA	REAL	4.0	Simulated signal (mA)
Out_A1	BOOL		Alarm level 1 active (to voting / beacons)
Out_A2	BOOL		Alarm level 2 active
Out_A3	BOOL		Alarm level 3 active
Out_Fault	BOOL		Detector fault / unavailable
Val_mA	REAL		Signal (mA)
Val_Conc	REAL		Gas concentration (Cfg_EU)
Val_Peak	REAL		Peak concentration since last reset
Sts_Inhibit	BOOL		Operator inhibit active
Sts_DetMaint	BOOL		Detector signalling maintenance
Sts_InhibitLeft	TIME		Inhibit time remaining
Sts_Sim	BOOL		Simulation active
Inp_NE107	INT	0	NE107 status reported by smart device (HART / field-bus)
Sts_NE107	INT		NAMUR NE107 status: 0=Good 1=Failure (F) 2=Function check (C) 3=Out of specification (S) 4=Maintenance required (M)
Alm_A1	T_AlmLim		Alarm level 1 (low) — set Cfg_Latch as required
Alm_A2	T_AlmLim		Alarm level 2 (high) — set Cfg_Latch as required
Alm_A3	T_AlmLim		Alarm level 3 (high-high / STEL)
Alm_Fault	T_Alarm		Detector fault
Alm_OverRng	T_Alarm		Over-range
Alm_InhibitLong	T_Alarm		Inhibit timed out

I.11.3 T_Intlk

Interlock / permissive group: 16 conditions, first-out, timed bypass.

ISA-88 role: Support (interlock group) **MTP:** LockView16 **Standards:** ISA-84 / IEC 61511 (BPCS layer only), ISA-18.2 / IEC 62682, ISA-101.01, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.32: T_Intlk Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Inp_Cond	ARRAY[0..15] OF BOOL		Condition n: TRUE = healthy / permitted
Cfg_Used	ARRAY[0..15] OF BOOL		Condition n is configured
Cfg_CondDesc	ARRAY[0..15] OF STRING[32]		HMI text for condition n
Cfg_Bypassable	ARRAY[0..15] OF BOOL		Condition n may be bypassed
Cfg_Latch	BOOL		Latch failures until Cmd_Reset (interlock); FALSE = permissive
Cfg_BypassMax	TIME	T#8H	Maximum bypass duration
Cmd_Bypass	ARRAY[0..15] OF BOOL		Bypass condition n (one-shot)
Cmd_Unbypass	ARRAY[0..15] OF BOOL		Remove bypass on condition n (one-shot)
Cmd_UnbypassAll	BOOL		Remove all bypasses (one-shot)
Inp_BypassPerm	BOOL		Bypass authorisation (key switch / HMI permit) — IEC 61511 bypass management
Cmd_Reset	BOOL		Reset latched conditions (one-shot)
Out_OK	BOOL		All used conditions healthy or bypassed — wire to Inp_IntlkOK / Inp_PermOK
Sts_OKNoByp	BOOL		All used conditions healthy without bypasses
Sts_Bypassed	ARRAY[0..15] OF BOOL		Condition n bypassed
Sts_Latched	ARRAY[0..15] OF BOOL		Condition n latched failed
Sts_FirstOut	INT		First condition to fail (-1 = none)
Sts_NumBypassed	INT		Number of bypassed conditions
Sts_BypassLeft	TIME		Time left on the oldest bypass
Sts_ResetReq	BOOL		Conditions healthy, reset required
Sts_CondWord	WORD		Packed Inp_Cond (bit n = condition n) for SCADA
Sts_BypWord	WORD		Packed Sts_Bypassed for SCADA
Sts_LatchWord	WORD		Packed Sts_Latched for SCADA
Alm_Trip	T_Alarm		Interlock tripped
Alm_Bypass	T_Alarm		One or more conditions bypassed

I.11.4 T_ESD

Emergency shutdown group (BPCS layer): de-energise-to-trip, MOS, first-out.

ISA-88 role: Support (shutdown group) **MTP:** - **Standards:** ISA-84 / IEC 61511 (BPCS layer only), ISA-18.2 / IEC 62682, ISA-101.01

Table I.33: T_ESD Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_Level	INT	1	ESD level, e.g. 0=Plant 1=Unit 2=Equipment
Inp_OK	ARRAY[0..15] OF BOOL		Trip input n: TRUE = healthy
Inp_UpstrOK	BOOL	TRUE	Higher-level ESD healthy (cascade)
Inp_ManTripOK	BOOL		Field ESD push-button circuit healthy
Inp_ResetPB	BOOL		Field reset push-button
Inp_MOSKey	BOOL		Maintenance override key switch enabled
Inp_FinalSafe	BOOL		All final elements confirmed in safe state
Cfg_Used	ARRAY[0..15] OF BOOL		Input n configured
Cfg_InDesc	ARRAY[0..15] OF STRING[32]		HMI text for input n
Cfg_MOSAllowed	ARRAY[0..15] OF BOOL		Override allowed on input n
Cfg_MOSMax	TIME	T#4H	Maximum override duration
Cfg_FieldReset	BOOL		Field reset push-button required to reset
Cfg_SafeTime	TIME	T#30S	Time for final elements to reach safe state
Cmd_ManTrip	BOOL		HMI manual trip (one-shot)
Cmd_Reset	BOOL		HMI reset (one-shot)
Cmd_MOS	ARRAY[0..15] OF BOOL		Apply override on input n (one-shot)
Cmd_MOSOff	ARRAY[0..15] OF BOOL		Remove override on input n (one-shot)
Out_Healthy	BOOL		TRUE = run permitted (energised); FALSE = trip
Sts_Tripped	BOOL		Group tripped
Sts_ResetReady	BOOL		All inputs healthy, reset possible
Sts_FirstOut	INT		First input to trip (-1 none, 16=manual, 17=upstream)
Sts_MOS	ARRAY[0..15] OF BOOL		Override active on input n
Sts_NumMOS	INT		Number of active overrides
Sts_InWord	WORD		Packed Inp_OK for SCADA
Sts_LatchWord	WORD		Packed latched trips for SCADA
Sts_MOSWord	WORD		Packed Sts_MOS for SCADA
Val_TripTime	DT		Date/time of last trip
Val_TripCount	UDINT		Trip counter
Alm_Trip	T_Alarm		ESD trip
Alm_FailSafe	T_Alarm		Final elements failed to reach safe state
Alm_MOS	T_Alarm		Maintenance override active
Alm_MOSLong	T_Alarm		Override exceeded Cfg_MOSMax

I.12 ISA-88 Batch and Procedural

I.12.1 T_Phase

ISA-88 equipment phase: Part 1 state model, commands, modes, parameters, reports.

ISA-88 role: Equipment phase (procedural) **MTP:** Service (state machine) **Standards:** ISA-88 / IEC 61512-1 Part 1 state model, VDI/VDE/NAMUR 2658-4 (MTP)

Table I.34: T_Phase Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_PhaseID	INT		Phase ID (unique in the unit) for batch server / unit procedure
Cfg_UnitID	INT		ID of the unit this phase belongs to
Cfg_StepTimeout	TIME	T#0S	Step watchdog, T#0S = off
Cmd_Auto	BOOL		Operator request Automatic mode (one-shot)
Cmd_SemiAuto	BOOL		Operator request Semi-automatic mode (one-shot)
Cmd_Man	BOOL		Operator request Manual mode (one-shot)
Cmd_Start	BOOL		Operator Start (one-shot)
Cmd_Stop	BOOL		Operator Stop (one-shot)
Cmd_Hold	BOOL		Operator Hold (one-shot)
Cmd_Restart	BOOL		Operator Restart from Held (one-shot)
Cmd_Abort	BOOL		Operator Abort (one-shot)
Cmd_Reset	BOOL		Operator Reset to Idle (one-shot)
Cmd_Pause	BOOL		Operator Pause at next transition (one-shot)
Cmd_Resume	BOOL		Operator Resume from Paused (one-shot)
Cmd_Advance	BOOL		Advance to next step in Semi-automatic (one-shot)
Cmd_TakeOwn	BOOL		Operator takes control from the recipe (one-shot)
Prog_Cmd	INT		Command from batch server / unit procedure: 1=Start 2=Stop 3=Hold 4=Restart 5=Abort 6=Reset 7=Pause 8=Resume
Prog_Acquire	BOOL		Recipe (batch server / unit procedure) takes ownership
Prog_Release	BOOL		Recipe releases ownership
Prog_ReqAck	BOOL		Batch server acknowledges Sts_ReqCode
Prm_Real	ARRAY[1..16] OF REAL		Recipe parameters (REAL) — control recipe formula
Prm_DInt	ARRAY[1..8] OF DINT		Recipe parameters (DINT) — control recipe formula
Rpt_Real	ARRAY[1..16] OF REAL		Report values (REAL) returned to batch record
Rpt_DInt	ARRAY[1..8] OF DINT		Report values (DINT) returned to batch record
Val_BatchID	STRING[32]		Batch ID of the control recipe running this phase
Val_StateTime	TIME		Time in current state
Val_RunTime	TIME		Running time excluding Held / Paused
Val_FailText	STRING[40]		Failure description for HMI / batch record
Sts_State	INT		ISA-88 state: 0=Idle 1=Running 2=Complete 3=Pausing 4=Paused 5=Holding 6=Held 7=Restarting 8=Stopping 9=Stopped 10=Aborting 11=Aborted
Sts_PrevState	INT		Previous state (for Restart / diagnostics)
Sts_Mode	INT		ISA-88 mode: 1=Manual 2=Semi-automatic 3=Automatic
Sts_Owner	INT		0=Operator 1=Recipe (batch server / unit procedure)
Sts_CmdMask	WORD		Commands valid in current state (bit n = command n of Prog_Cmd) — for HMI

Table I.34 continued

Member	Type	Init	Description
Sts_CmdRejected	BOOL		Last command was not valid in the current state
Sts_Step	INT		Active step number within the state logic
Sts_WaitAdvance	BOOL		Semi-automatic: waiting for Cmd_Advance
Sts_ReqCode	INT		Request to batch server (e.g. download params, upload reports), 0 = none
Sts_FailCode	INT		Failure code, 0 = none
Alm_Fail	T_Alarm		Phase failure
Alm_StepTimeout	T_Alarm		Step exceeded Cfg_StepTimeout

I.12.2 T_Unit

ISA-88 unit: allocation, batch identity, procedural and equipment state.

ISA-88 role: Unit (physical) **MTP:** - **Standards:** ISA-88 / IEC 61512-1, ISA-18.2 / IEC 62682, ISA-101.01

Table I.35: T_Unit Members

Member	Type	Init	Description
Cfg_Tag	STRING[20]		Equipment / instrument tag, e.g. FT-101
Cfg_Desc	STRING[40]		Description shown on HMI faceplate
Cfg_UnitID	INT		Unit ID
Cfg_UnitClass	STRING[24]		Unit class name (master recipe equipment requirement)
Cfg_CleanHold	TIME	T#72H	Clean hold time — clean status expires after this
Inp_ReqOwnerID	DINT		ID of control recipe requesting allocation
Prog_Acquire	BOOL		Allocate unit to requesting control recipe
Prog_Release	BOOL		Release unit
Cmd_ForceRelease	BOOL		Supervisor forces release (one-shot)
Prog_EqState	INT		Equipment state request from phase logic
Cmd_EqState	INT		Operator equipment state change (e.g. to Maintenance)
Val_OwnerID	DINT		Allocated control recipe ID, 0 = free
Val_BatchID	STRING[32]		Batch ID
Val_RecipeName	STRING[32]		Control recipe / master recipe name
Val_RecipeVer	INT		Recipe version
Val_ProductCode	STRING[24]		Product / material code
Val_BatchSize	REAL		Batch size (scaled)
Val_BatchStart	DT		Batch start date/time
Val_LastClean	DT		Date/time of last completed clean
Sts_Allocated	BOOL		Unit allocated to a control recipe
Sts_ProcState	INT		Unit procedure state: 0=Idle 1=Running 2=Complete 3=Pausing 4=Paused 5=Holding 6=Held 7=Restarting 8=Stopping 9=Stopped 10=Aborting 11=Aborted
Sts_Mode	INT		Unit procedure mode: 1=Manual 2=Semi-automatic 3=Automatic
Sts_ActivePhase	INT		Cfg_PhaseID of the active phase, 0 = none
Sts_EqState	INT		ISA-88 equipment state: 0=Available-clean 1=In use 2=Dirty 3=Cleaning (CIP) 4=Sterilising (SIP) 5=Maintenance 6=Out of service
Sts_CleanExpired	BOOL		Clean hold time exceeded

Table I.35 continued

Member	Type	Init	Description
Alm_CleanExpired	T_Alarm		Clean status expired

I.13 Reference Implementation: FB_AI

FB_AI is the first PCLib DFB: an IEC 61131-3 Structured Text block that processes one T_AI instance. It is generated by gen_dfb.py and imported from export/PCLib_DFB_AI_CE_V16.2.xdb, which also contains the shared alarm blocks FB_AlmCore, FB_Alarm, and FB_AlmLim. The same alarm blocks are intended to be reused by every other PCLib DFB as it is written. The block uses no located variables, global variables, or system bits, so all data passes through its parameters. Run it once per scan in a cyclic task; it measures its own elapsed time.

```
(* Variables: FT101 : T_AI;  fbFT101 : FB_AI; *)
FT101.Inp_Raw     := %IW0.2.0;            (* or the M580 channel value *)
FT101.Inp_ChFault := NOT %I0.2.0.ERR;     (* channel health *)
fbFT101(AI := FT101);
```

Table I.36: FB_AI Processing Order

#	Step	Members
1	Elapsed time since last scan (free-running TON, no system calls)	-
2	Configuration check: raw or EU span of zero makes the PV bad	Cfg_RawMin/Max, Cfg_EUMin/Max
3	Raw counts to % of span	Inp_Raw
4	Signal validation: NAMUR NE43 limits, channel fault, device NE107 Failure	Cfg_BadLoPct/HiPct, Inp_ChFault, Inp_NE107, Sts_URng/ORng
5	Square-root extraction, low cut-off, scaling to EU	Cfg_SqrtExtract, Cfg_LoCutoff
6	Simulation replaces the field signal, and field faults are ignored	Sim_En, Sim_PV, Sts_Sim
7	First-order filter. Re-initialised on start-up, bad-to-good recovery and sim change, so there is no bump	Cfg_FiltTC
8	Bad-value action: hold last good / substitute / EUMin / EUMax	Cfg_BadAction, Cfg_SubstPV, Sts_Subst
9	Quality and NAMUR NE107 status. Simulation reports as C (function check)	Sts_Quality, Sts_NE107
10	Rate of change, 1 s window. Re-initialised on recovery and sim change, so there are no false ROC alarms	Val_ROC
11	Alarms: HH/H/L/LL/ROC limit alarms and the bad-PV alarm	Alm_*

Limit alarms are disabled by default (Cfg_En = FALSE); enable each one after its limit has been set in alarm rationalisation. The simulator test checklist for FB_AI is kept with the library in the EFC Repo.